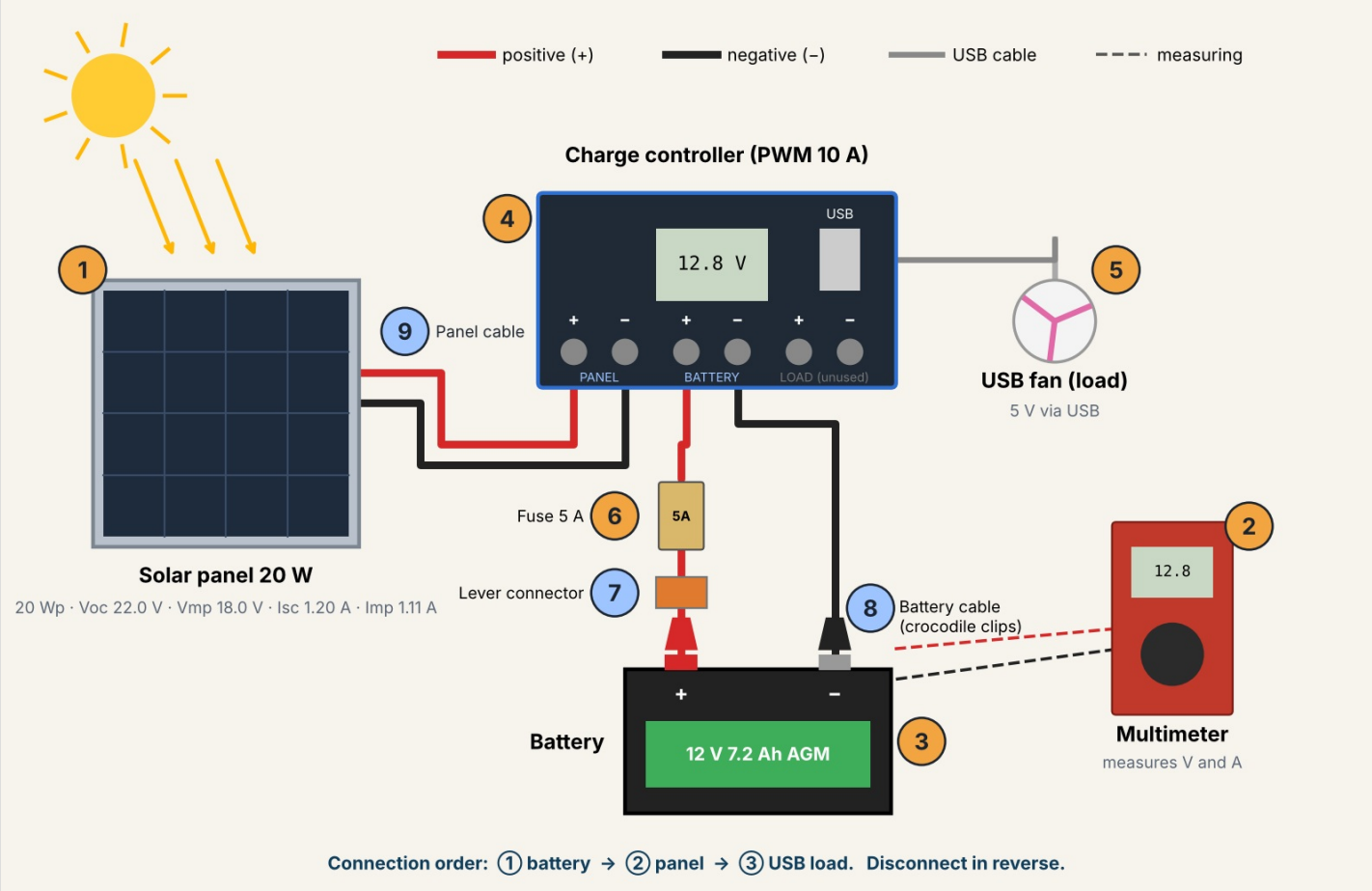
Solar + Battery Test System — Readings Log

Mohamed Eniw · Passau, Germany · October 2026 · values measured with an AstroAI AM33D multimeter

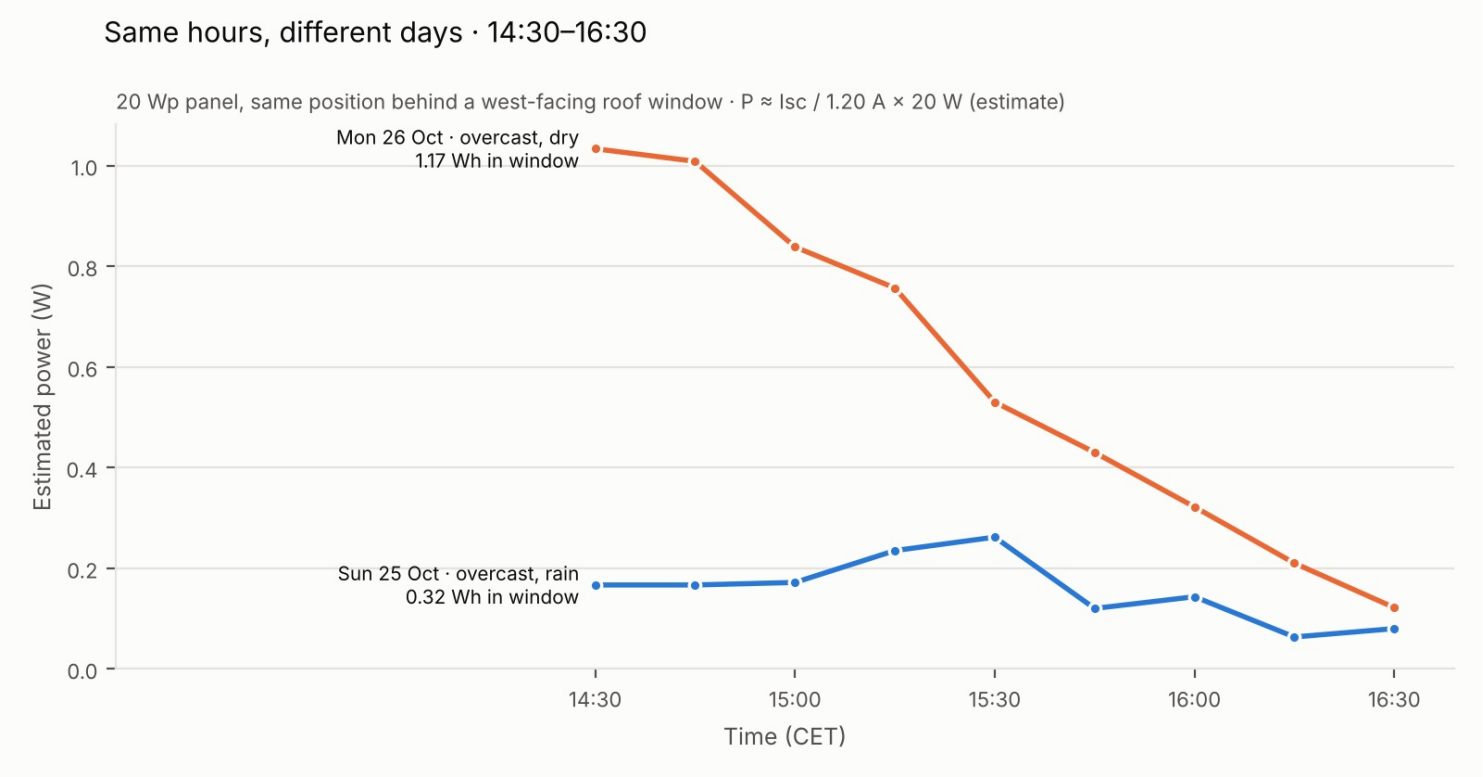
#	Measurement	Parts	Meter setting	Expected	Measured	Time / conditions
R1	Battery voltage at rest (before wiring)	2 on 3	V⎓ 20	12.6–13.0 V	**12.94 V**	Sat 24 Oct, 18:01, indoors
R2	Battery voltage shown on controller	4	display	≈ R1	**12.7 V**	Sat 24 Oct, 19:22, indoors
R3	Battery voltage with fan running	2 on 3	V⎓ 20	slightly below R1	**12.68 V**	Sat 24 Oct, 19:29, indoors
R4	Panel open-circuit voltage (Voc), sun	2 on 1	V⎓ 200	≈ 20–22.0 V (label Voc 22.0 V)	**21.8 V**	Sun 25 Oct, 10:54, behind window, overcast/rain
R5	Panel short-circuit current (Isc), sun	2 on 1	A 10 (10 A socket)	≤ 1.20 A (label Isc)	**0.04 A**	Sun 25 Oct, 10:59, behind window, overcast/rain
R6	Battery voltage while charging	2 on 3	V⎓ 20	rises above R1		
R7	Charging current panel → controller	2 in 9	A 10 (10 A socket)	up to ≈ 1 A		

Parts: 1 Solar panel 20 W · 2 Multimeter · 3 Battery 12 V 7.2 Ah AGM · 4 Charge controller PWM 10 A · 5 USB fan · 6 Fuse 5 A · 7 Lever connector · 8 Battery cable · 9 Panel cable



Daily production log

Short-circuit current (Isc) logged by camera every 15 min · panel fixed behind a west-facing roof window (284°) · P ≈ Isc / 1.20 A × 20 W (estimate)



Day	Weather	Peak	Energy (logged window)
Sun 25 Oct	Overcast, rain (high 10 °C)	0.05 A at 12:45 (≈ 0.83 W, ≈ 42 W/m²)	≈ 2.0 Wh (11:15–16:30)
Mon 26 Oct	Overcast, dry (high 12 °C)	62 mA at 14:30 (≈ 1.03 W, ≈ 52 W/m²)	≈ 1.2 Wh (14:30–17:00)
Tue 27 Oct	–	–	–